Exploring Two-Step Geographic Aggregation via Nested Redistricting

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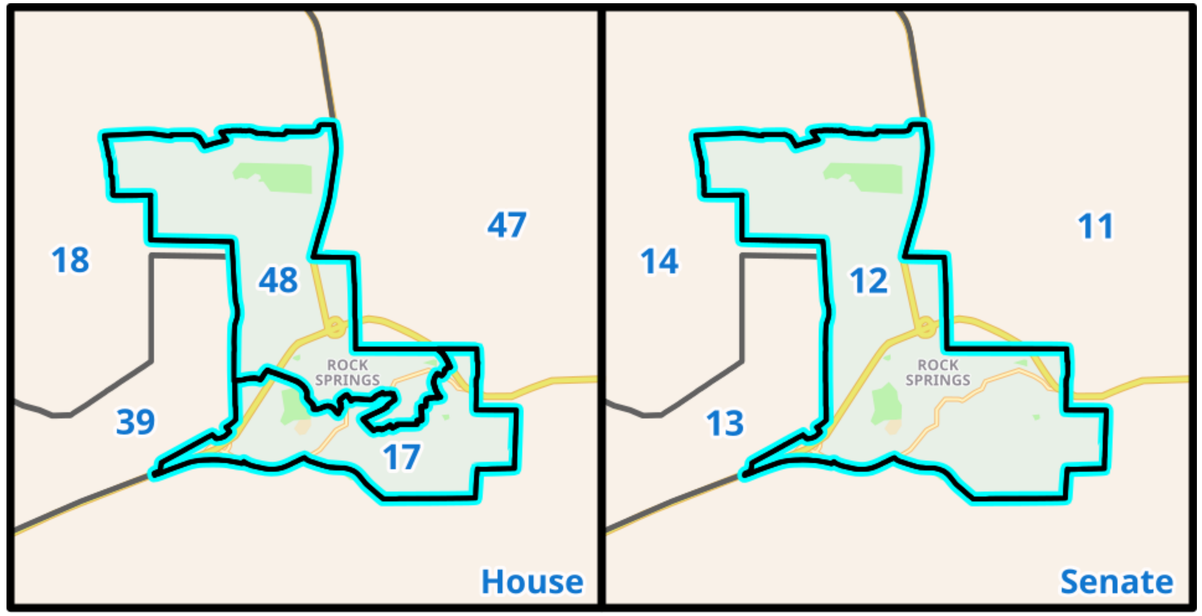
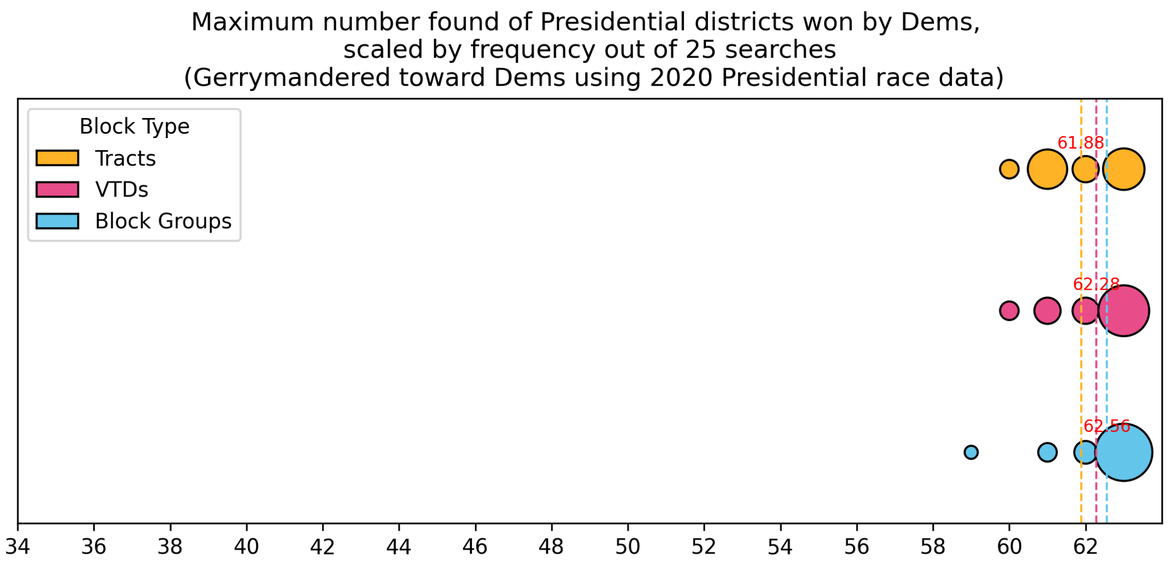
- **Nested redistricting**: a strategy of building districts for one body by aggregating the districts of another body.

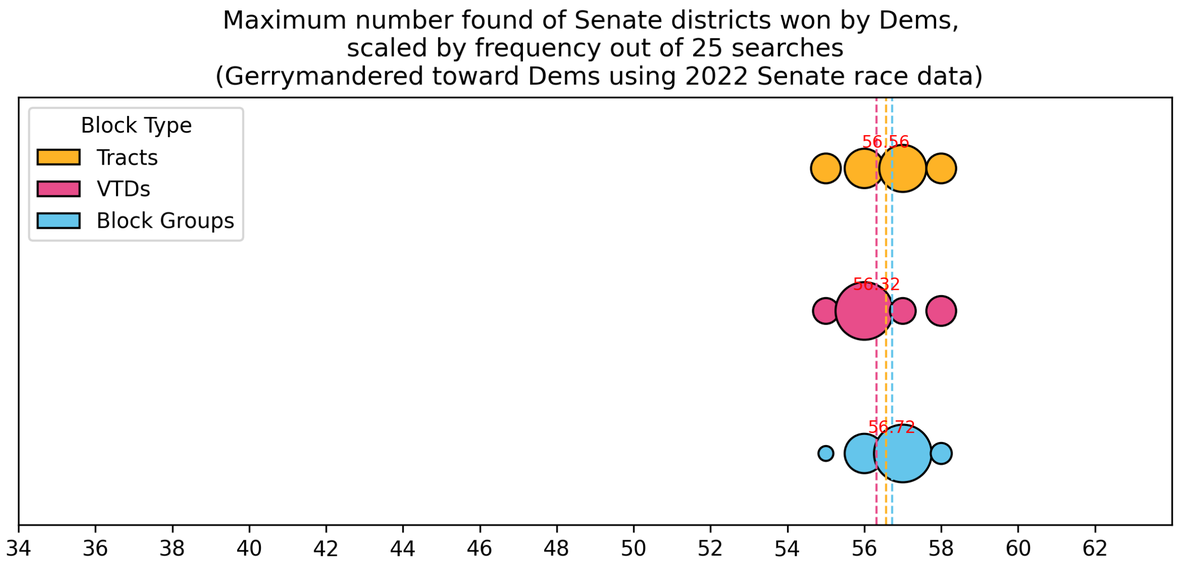
- **Guiding question:** how does the size of the pieces being aggregated affect the freedom of the agent building the districts, particularly with regard to the ability to gerrymander?



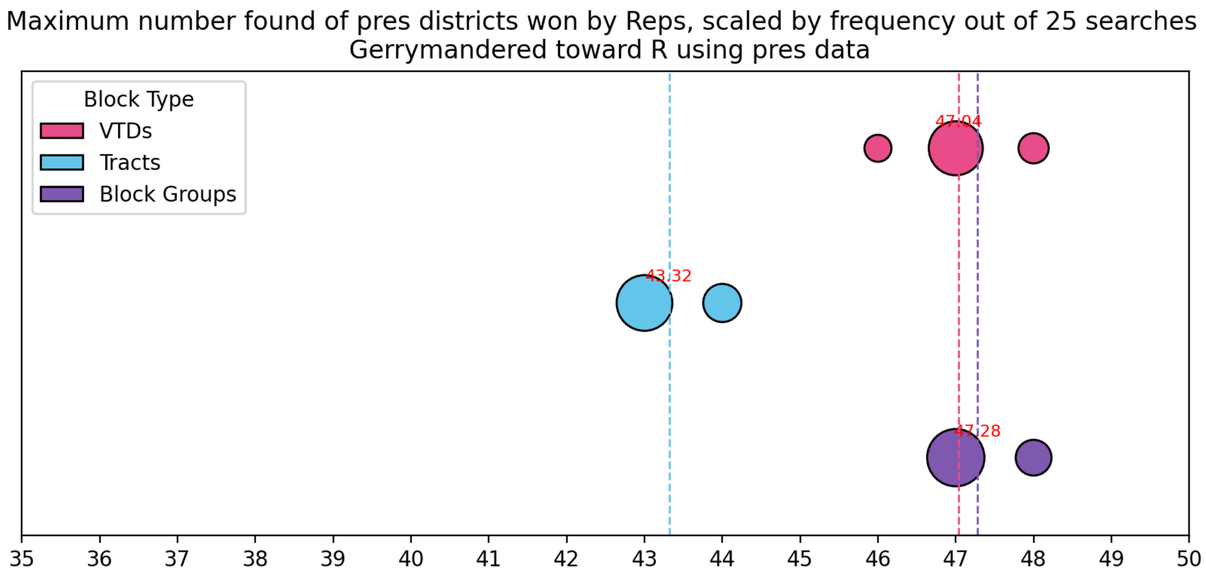
via Wikimedia Commons

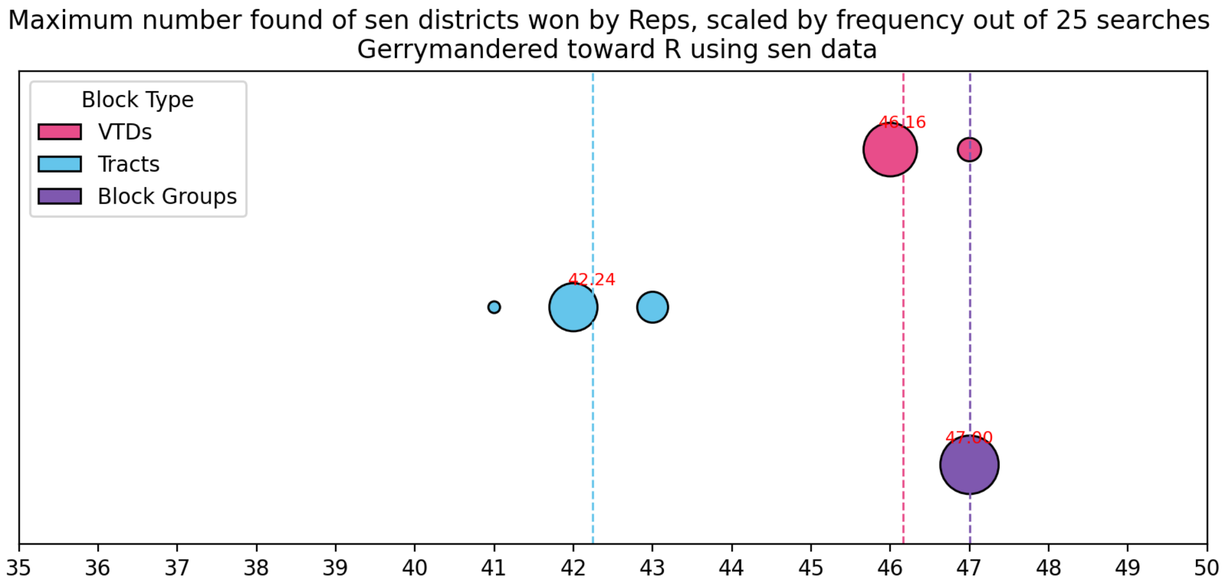
In New York:


Maximum number found of Presidential districts won by Dems,
scaled by frequency out of 25 searches
(Gerrymandered toward Dems using 2020 Presidential race data)
Block Type
Tracts
VTDs
Block Groups
61.88
62.28
62.56
34
36
38
40
42
44
46
48
50
52
54
56
58
60
62

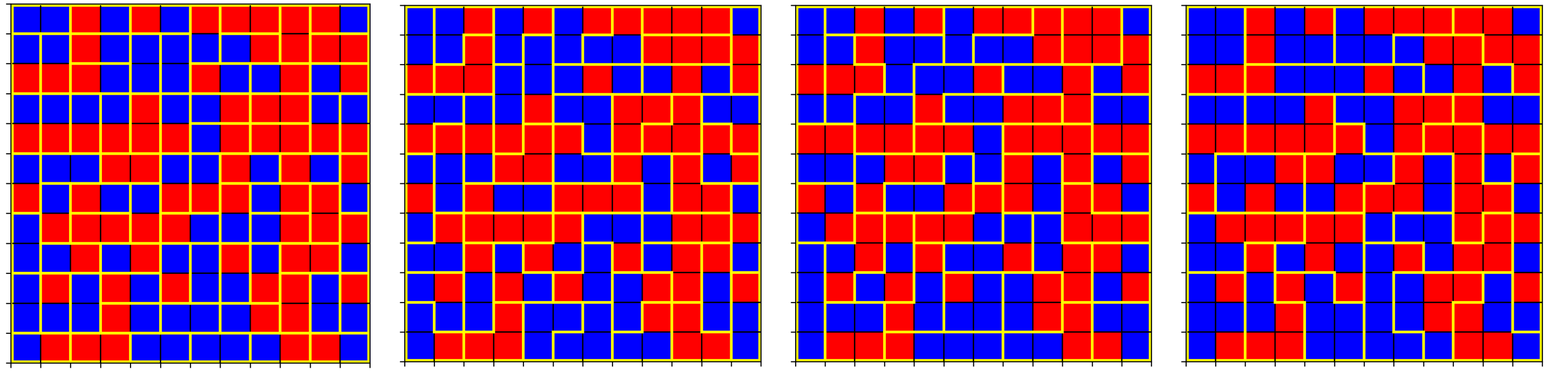
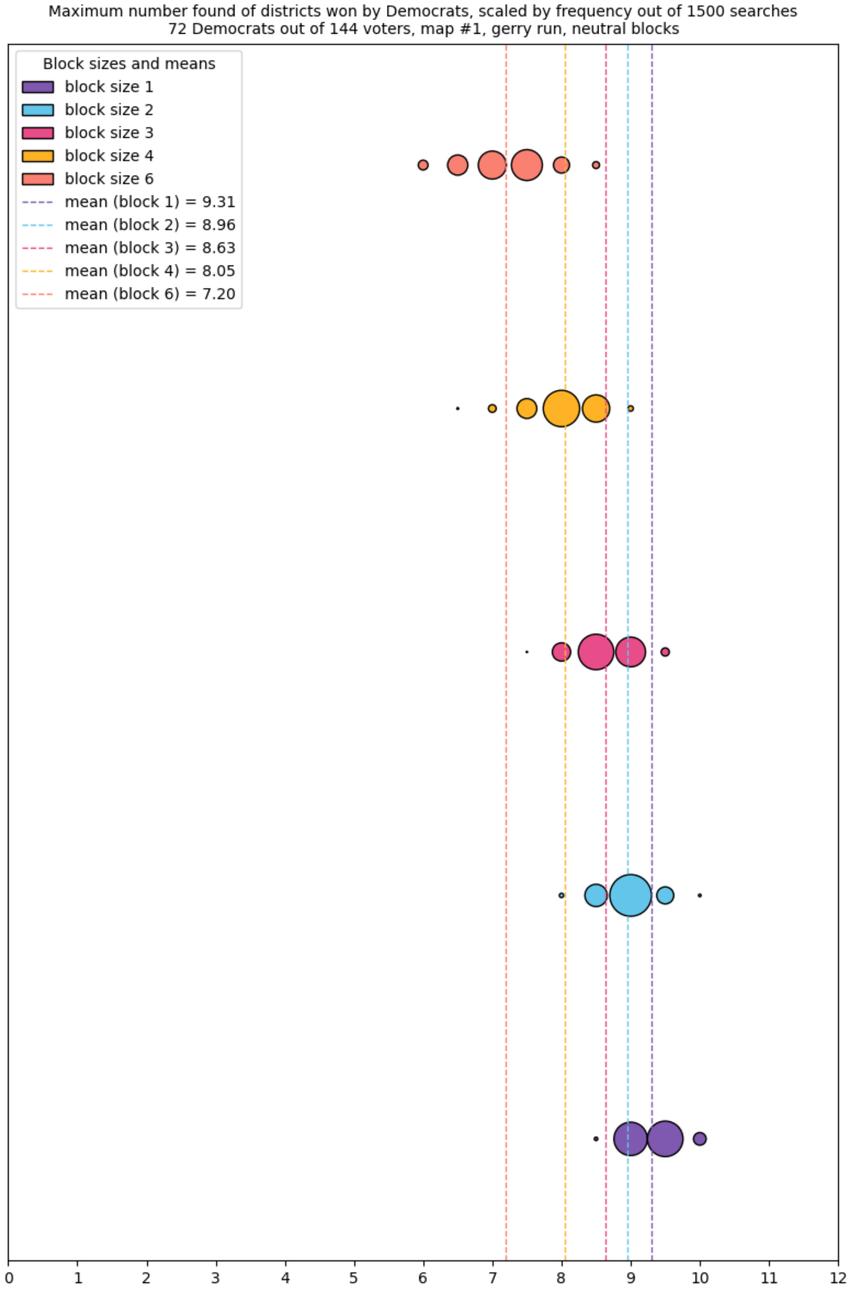

Maximum number found of Senate districts won by Dems,
scaled by frequency out of 25 searches
(Gerrymandered toward Dems using 2022 Senate race data)
Block Type
Tracts
VTDs
Block Groups
56.56
56.32
56.72
34
36
38
40
42
44
46
48
50
52
54
56
58
60
62

But in Montana…





What about in the grid?

- Shift view to a 12x12 grid, divided into building blocks of sizes 1, 2, 3, 4, or 6, which will be combined into 12 districts.

- **New question:** in this setting, would you rather control the drawing of the districts or of the blocks? If you control the drawing of the blocks, what is the best strategy to maximize return at the level of districts?